

SPARSITY-BASED IMAGE DEBLURRING WITH LOCALLY ADAPTIVE AND NONLOCALLY ROBUST REGULARIZATION

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ABSTRACT

Important structures in photographic images such as edges and textures are jointly characterized by local variation and nonlocal invariance (similarity). Both of them provide valuable heuristics to the regularization of image restoration process. In this paper, we propose to explore two sets of complementary ideas: 1) locally learn PCA-based dictionaries and estimate the sparsity regularization parameters for each coefficient; and 2) nonlocally enforce the invariance constraint by introducing a patch-similarity based term into the cost functional. The minimization of this new cost functional leads to an iterative thresholding-based image deblurring algorithm and its efficient implementation is discussed. Our experimental results have shown that the proposed scheme significantly outperforms several leading deblurring techniques in the literature on both objective and visual quality assessments.

Index Terms— Image deblurring, iterative shrinkage, sparsity-based local adaptation, nonlocal similarity

I. INTRODUCTION

The basic idea behind sparse image representations is to assume that image signal $\mathbf{x}$ can be well approximated by a small set of atoms from a dictionary $\mathbf{D} \in R^{n \times m}$ - i.e., $\mathbf{x} = \mathbf{D}\alpha$ where α is expected to be sparse. Image deblurring - the estimation of $\mathbf{x}$ from its noisy blurred version $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{w}$, $\mathbf{w} \sim N(0, \sigma_w^2)$ - can be casted as an optimization problem as follows:

$$\hat{\alpha} = \arg \min_{\alpha} \{ \|\mathbf{y} - \mathbf{H}\mathbf{D}\alpha\|_2^2 + \lambda \|\alpha\|_{l_0} \}, \quad (1)$$

where the l_0 -norm denotes the number of nonzero entries in a vector. Under this sparsity-based framework, several lines of ideas have developed rapidly in the recent years. First, there is a trend of learning dictionary $\mathbf{D}$ on-the-fly instead of adopting a fixed one [1], [2]. It has been well observed that a redundant dictionary ($n \ll m$) could lead to higher sparsity and therefore beneficial to signal recovery. Second, since the problem of l_0 -optimization is NP-hard, it is often replaced by its l_1 -counterpart, which can be solved efficiently by convex optimization techniques. Moreover, the idea of reweighted l_1 -optimization [3], which achieves spatial adaptation by locally adjusting the λ parameter, has been shown promising in compressed sensing experiments. Third, it has been suggested that structural clustering could give rise to so-called group sparsity [4], [5]. The exploitation of nonlocal similarity in $\mathbf{x}$ has been shown particularly effective for the class of image structures such as regular edges and textures.

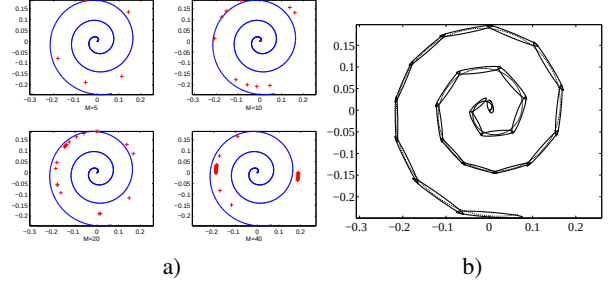


Fig. 1. Two ways of discovering the manifold structure: a) dictionary learning (atoms in $\mathbf{D}$ are highlighted in red '+'); b) structural clustering (each segment denotes a different cluster $p(\mathbf{x}|\theta_m)$).

It is enlightening to understand and compare the above approaches using a geometric tool called manifold learning [6]. The collection of image patches with the same size $B \times B$ forms a low-dimensional manifold $\mathcal{M}$ such that $\dim(\mathcal{M}) \ll B^2$ [7]. Various sparsity-based image priors represent competing views toward the characterization of such manifold constraint. To facilitate our discussion, let's consider a toy example of spiral data (1D manifold embedded in a 2D space) as shown in Fig. 1a. Dictionary learning such as K-SVD algorithm attempts to approximate the spiral structure by putting atoms along all different directions; its capability of manifold learning appears to be strictly limited by the linear structure of $\mathbf{x} = \mathbf{D}\alpha$. By contrast, structural clustering by the EM-algorithm [8] decomposes the manifold into the mixture of finite segments/models $p(\mathbf{x}|\theta) = \sum_{m=1}^k \alpha_m p(\mathbf{x}|\theta_m)$ (refer to Fig. 1b). The formulation of reweighting can also be connected with the concept of clustering because varying λ 's essentially reflects the idea of classifying a nonstationary source. The subtle difference lies in the clustering heuristics: pixels could be clustered into the class either because they are locally adjacent and nonlocally correlated (i.e., adjacent in the patch space).

In this paper, we propose to combine the idea of dictionary learning with structural clustering and demonstrate its application into image deblurring applications. Our approach can be formulated under a variational framework - the key innovation lies in the introduction of a new regularization functional which attempts to strike a good balance between *local adaptation* and *nonlocal robustness*. The local term in our regularization functional is inspired by the well-known results on statistical modeling of wavelet coefficients [9]. We show how the regularization parameter λ can be

made spatially adaptive by estimating the local variance of image signals. The nonlocal term is inspired by the idea of nonlocal-mean denoising [10] but ours is formulated in the transform domain, which makes it convenient to be incorporated into the likelihood term. We show how to solve the formulated variational problem by iterative shrinkage techniques; however, ours update both learned dictionary and regularization parameters on-the-fly. Geometrically our algorithm can be interpreted as manifold learning via *hybrid* heuristics - both local geometry (i.e., the correlation dependency of projected subspaces arising from the local smoothness in the spatial domain) and global topology (i.e., the neighborhood relationship in the patch space arising from the nonlocal similarity in the spatial domain) contribute to the inference of individual pixels. Experimental results have shown that the proposed deblurring algorithm significantly outperforms several leading techniques in the literature.

II. LOCALLY ADAPTIVE AND NONLOCALLY ROBUST REGULARIZATION

Let us start from some notations. For an image $\mathbf{x} \in R^{H \times W}$, we consider its patch-based representation $\mathbf{x}_i = \mathbf{R}\mathbf{x} \in R^n, i = 1, 2, \dots, N$ (n is patch size) where $\mathbf{R}$ denotes an operator extracting patch $\mathbf{x}_i$ from $\mathbf{x}$. Assuming a given dictionary $\mathbf{D}_i$ for $\mathbf{x}_i$, we have $\mathbf{x}_i = \mathbf{D}_i \alpha_i$ which implies that $\{\alpha_i\}$ is a sparse and redundant representation of $\mathbf{x}$. More specifically, we can write the reconstruction formula as [2]

$$\mathbf{X} = \left(\sum_{i=1}^N \mathbf{R}_i^T \mathbf{R}_i \right)^{-1} \left(\sum_i \mathbf{R}_i^T \mathbf{x}_i \right), \quad (2)$$

Then the problem of image deblurring can be formulated into the following baseline optimization problem

$$\hat{\alpha} = \arg \min_{\alpha} \{ \|\mathbf{y} - \mathbf{H}\mathbf{D}\alpha\|_{l_2}^2 + \lambda \sum_{i=1}^N \|\alpha_i\|_{l_1} \}, \quad (3)$$

The iterative shrinkage solution to the above l_1 -optimization has been studied extensively in the literature (e.g., surrogate function [11]). Regarding the choice of dictionary $\mathbf{D}_i$. One could choose a fixed set of atoms (e.g., DCT or wavelet bases) or learn the dictionary from the partially restored image $\hat{\mathbf{x}}$. For example, our previous work on adaptive PCA-based image denoising has shown how to apply PCA to a collection of similar patches found both within the same scale and across different scales. As the estimation of $\hat{\mathbf{x}}$ gets updated at each iteration, a new dictionary can be learned on-the-fly (more details can be found in the recent work [12]).

Note that a constant regularization parameter λ implies the stationarity assumption about the image source. It is a natural idea to introduce spatially adaptive regularization to accommodate the nonstationary nature of photographic images. More specifically, we can consider

$$\hat{\alpha} = \arg \min_{\alpha} \{ \|\mathbf{y} - \mathbf{H}\mathbf{D}\alpha\|_{l_2}^2 + \sum_{i=1}^N \sum_{j=1}^m \lambda_{i,j} |\alpha_{i,j}| \}, \quad (4)$$

where $\lambda_{i,j}$ denotes the regularization parameter for the i -th coefficient (spatial index) in the j -th atom (frequency index) in the dictionary (i.e., $\mathbf{D}_j$). In the work of reweighted l_1 -regularization [3], the weight $\lambda_{i,j} = \frac{1}{|\hat{\alpha}_{i,j}| + \epsilon}$ is used. In our recent work [13], we have developed a Bayesian approach toward spatial adaptation

(conceptually related to Bayesian modeling of wavelet coefficients [9]) which gives

$$\lambda_{i,j} = \frac{2\sqrt{2}\sigma_w^2}{|\hat{\sigma}_{i,j}| + \epsilon}, \quad (5)$$

where $\sigma_{i,j}, \sigma_w$ denote the variance of signal and noise respectively (we will elaborate on the estimation of σ_x later since it involves nonlocal neighborhood as well). It is not surprising to observe that the solution to Eq. (4) is given by iterative shrinkage with spatially adaptive thresholds ($\tau_{i,j} = \frac{\lambda_{i,j}}{r}$ where r is a constant) [14].

Now let us consider the inclusion of nonlocal regularization term. For any patch $\mathbf{x}_i$, let us assume its k -nearest-neighbors (kNN) in the patch space are $\mathbf{x}_i^{(1)}, \dots, \mathbf{x}_i^{(k)}$. Inspired by the nonlocal-mean denoising [10], we can define the following weighted prediction

$$\hat{\mathbf{x}}_i = \sum_{t=1}^k w_{i,t} \mathbf{x}_i^{(t)}, w_{i,t} \sim \exp(-\|\mathbf{x}_i - \mathbf{x}_i^{(t)}\|/h) \quad (6)$$

where h is a constant controlling the window shape. It follows from Eq. (6) that the nonlocal similarity in an image can be approximately characterized by

$$J_{NL}(\mathbf{x}) = \sum_{i=1}^N \left\| \mathbf{x}_i - \sum_{t=1}^k w_{i,t} \mathbf{x}_i^{(t)} \right\|_{l_2}^2, \quad (7)$$

which can be written more compactly in a matrix form

$$J_{NL}(\mathbf{x}) = \sum_{i=1}^N \|(\mathbf{I} - \mathbf{W})\mathbf{x}_i\|_{l_2}^2 = \|(\mathbf{I} - \mathbf{W})\mathbf{D}\alpha\|_{l_2}^2, \quad (8)$$

where $\mathbf{I}, \mathbf{W} = [w_{i,t}]$ denote the identity and NL-similarity matrix respectively. Putting Eqs. (4) and (8) together, we have the following new regularization functional

$$\hat{\alpha} = \arg \min_{\alpha} \{ \|\mathbf{y} - \mathbf{H}\mathbf{D}\alpha\|_{l_2}^2 + \sum_{i=1}^N \sum_{j=1}^m \lambda_{i,j} |\alpha_{i,j}| \} + \eta \|(\mathbf{I} - \mathbf{W})\mathbf{D}\alpha\|_{l_2}^2. \quad (9)$$

where a new regularization parameter η controls the balance between local adaptation and nonlocal robustness. When compared with Eq. (4), we note that the inclusion of nonlocal regularization term not only directly enforces nonlocal similarity but also indirectly facilitates the estimation of signal variance required by $\lambda_{i,j}$.

III. NEW IMAGE DEBLURRING ALGORITHM

At the first sight, solving Eq. (9) seems a daunting task because of the newly added nonlocal regularization term. In fact, we can “merge” this term into the likelihood term and reduce Eq. (9) into a form similar to Eq. (4). Specifically, we can define

$$\tilde{\mathbf{y}} = \begin{bmatrix} \mathbf{y} \\ 0 \end{bmatrix}, \mathbf{K} = \begin{bmatrix} \mathbf{H} \\ \eta(\mathbf{I} - \mathbf{B}) \end{bmatrix}. \quad (10)$$

and rewrite Eq. (9) into

$$\hat{\alpha} = \arg \min_{\alpha} \{ \|\tilde{\mathbf{y}} - \mathbf{K}\mathbf{D}\alpha\|_{l_2}^2 + \sum_{i=1}^N \sum_{j=1}^m \lambda_{i,j} |\alpha_{i,j}| \}. \quad (11)$$

So we can see iterative shrinkage solution to Eq. (4) is still applicable as long as we replace $(\mathbf{y}, \mathbf{H})$ by $(\tilde{\mathbf{y}}, \mathbf{K})$. This leads to a complete image deblurring algorithm as follows.

Algorithm 1. Image Deblurring with Locally Adaptive and Nonlocally Robust Regularization (LANL)

Input: noisy blurred $\mathbf{y}$ and blurring kernel $\mathbf{H}$

Output: estimated image $\hat{\mathbf{x}}$

• Initialization:

- obtain $\hat{\mathbf{x}}^{(0)}$ by solving Eq. (2) with $\mathbf{D}$ being DCT bases;
- Using the initial estimate $\hat{\mathbf{x}}^{(0)}$ to calculate PCA bases $\mathbf{D}_i$, local sparsity regularization parameters $\lambda_{i,j}$ via Eq. (5 and nonlocal weight matrix $\mathbf{W}$ using Eq. (6);

• Main loop: Iterate until $\|\hat{\mathbf{x}}^{(k+1)} - \hat{\mathbf{x}}^{(k)}\|_{l_2}^2 \leq \epsilon$ or for $k \geq k_{max}$,

- Landweber iteration: $\hat{\mathbf{x}}^{(k+1/2)} = \hat{\mathbf{x}}^{(k)} + \mathbf{K}^T(\tilde{\mathbf{y}} - \mathbf{K}\hat{\mathbf{x}}^{(k)})$;
- PCA: $\alpha^{(k+1/2)} = [\mathbf{D}_1^T \mathbf{R}_1 \hat{\mathbf{x}}^{(k+1/2)}, \dots, \mathbf{D}_N^T \mathbf{R}_N \hat{\mathbf{x}}^{(k+1/2)}]$;
- Soft Thresholding: $\alpha^{(k+1)} = Th(\alpha^{(k+1/2)}, \tau_{i,j})$ where $Th(\cdot, \tau)$ denotes a soft thresholding operator with threshold τ ;
- Image reconstruction: $\hat{\mathbf{x}}_i^{(k+1)} = \mathbf{D}_i \alpha_i^{(k+1)}$ and obtain $\hat{\mathbf{x}}^{(k+1)}$

by Eq. (2);

- Dictionary updating: recalculate $\mathbf{D}_i$, $\lambda_{i,j}$ and $\mathbf{W}$ using $\hat{\mathbf{x}}^{(k+1)}$.

Next, we discuss some further implementation details. Substituting Eq. (10) into Landweber iteration, we obtain

$$\hat{\mathbf{x}}^{(k+1/2)} = \hat{\mathbf{x}}^{(k)} + \mathbf{H}^T(\mathbf{y} - \mathbf{H}\hat{\mathbf{x}}^{(k)}) - \mathbf{V}\hat{\mathbf{x}}^{(k)}. \quad (12)$$

where $\mathbf{V} = \eta^2(\mathbf{I} - \mathbf{B})^T(\mathbf{I} - \mathbf{B})$. Therefore, Eq. (12) can be conveniently implemented by a Landweber iteration followed by nonlocal-mean subtraction. The thresholds $\tau_{i,j}$ are locally calculated by $\tau_{i,j} = \frac{\lambda_{i,j}}{r}$ where r is chosen such that $r > \|(\mathbf{K}\mathbf{D})^T(\mathbf{K}\mathbf{D})\|_{l_2}$ (convergence condition of Landweber iteration). From Eq. (5), we can see that $\lambda_{i,j}$ is connected with signal variance $\sigma_{i,j}$. Note that due the presence of nonlocal similar patches, the estimation of $\sigma_{i,j}$ can be based on the neighborhood in both spatial domain and patch space. To further tailor down the complexity, we only apply the PCA-based dictionary learning to patches whose variance is above a chosen threshold (i.e., likely to be edges and textures); for patches in the smooth areas, we treat them as a whole class and only compute their PCA transformation matrix once. The complete package including MATLAB codes and saved experimental results can be downloaded from <http://www.csee.wvu.edu/~xin/code/ASDNL.rar>.

IV. EXPERIMENTAL RESULTS

We have compared our deblurring technique with five latest published works: 1) fast iterative shrinkage/thresholding algorithm (FISTA) [15]; 2) spatially-weighted total-variation (SWTV) [16]; 3) l_0 -sparsity based deblurring [17]; 4) shape-adaptive DCT (SA-DCT) deblurring [18] and 5) BM3D-based deblurring [19]. The experimental results of all benchmark schemes are obtained from the source codes or executables provided by their authors. Due to space limitation¹, we will only report the PSNR/SSIM comparison results for 9×9 uniform blur kernel and noise level of $\sigma_w = \sqrt{2}$ here. It can be observed from Table I that our LANL approach outperforms the competing ones in most situations except that BM3D-deblurring works better on *barbara* image. This is not surprising because BM3D emphasizes more on exploiting nonlocal similarity; while our approach attempts to strike a good balance between local variation and nonlocal invariance. Fig. 2 includes the subjective quality comparison (l_0 -sparsity was not shown because

it appears noticeably inferior to others). Visual inspection shows that our approach is the most preferred one because it works well for both the class of edge and texture structures. For *parrot* image containing both edge (e.g., beak of the parrot) and texture (e.g., fur pattern around the eye) structures, FISTA and SWTV noticeably fall behind because their recovered images look less natural; SA-DCT and BM3D did a better job but their recovered fine-detailed structures are not as sharp as that by LANL. This specific example clearly justifies the importance of jointly exploiting local adaptation and nonlocal robustness.

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¹More experimental results can be found in the software package accompanying this paper.

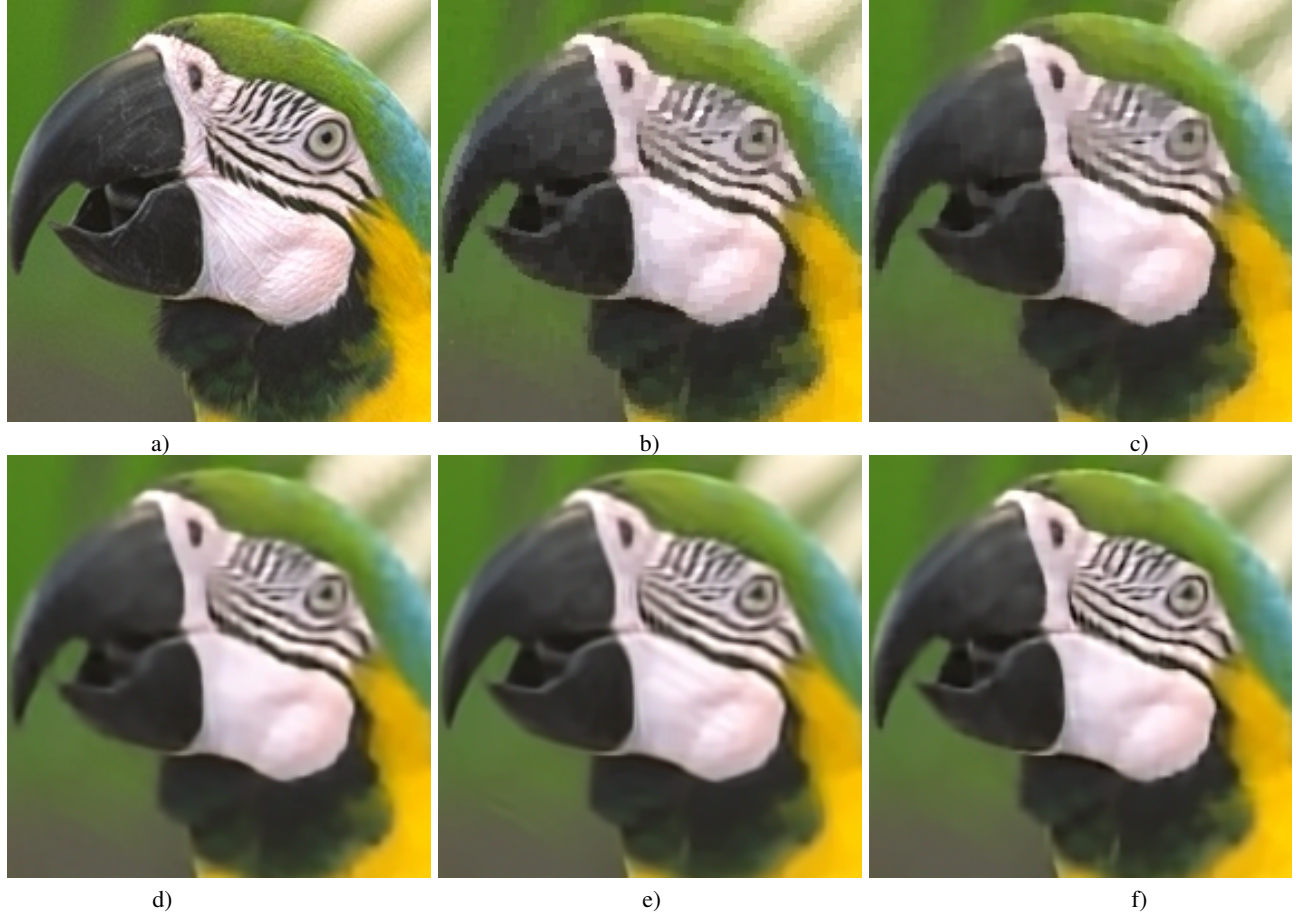


Fig. 2. Subjective quality comparison among deblurred images by various algorithms: a) original *parrot*; b) FISTA [15] ($PSNR = 28.80dB$, $SSIM = 0.8704$); c) SWTV [16] ($PSNR = 28.96dB$, $SSIM = 0.8722$); d) SA-DCT [18] ($PSNR = 29.78dB$, $SSIM = 0.8789$); e) BM3D [19] ($PSNR = 30.22dB$, $SSIM = 0.8906$); f) Ours ($PSNR = 31.33dB$, $SSIM = 0.9039$).

Table I. PSNR (dB) and SSIM results of various deblurring algorithms (9×9 uniform blur kernel, $\sigma_w = \sqrt{2}$).

images	FISTA [15]	SWTV [16]	l_0 Sparsity [17]	SA-DCT [18]	BM3D [19]	Ours
Parrot	28.80	28.96	29.04	29.78	30.22	31.33
	0.8704	0.8722	0.8824	0.8789	0.8906	0.9039
Butterfly	28.40	27.96	27.12	27.54	27.22	29.15
	0.9075	0.8897	0.8902	0.8939	0.8910	0.9208
Cameraman	26.72	27.38	26.86	26.98	27.24	28.01
	0.8330	0.8443	0.8361	0.8222	0.8308	0.8636
Lena	28.14	28.73	28.52	28.27	28.80	29.47
	0.8233	0.8432	0.8423	0.8255	0.8380	0.8616
Peppers	28.44	28.87	28.75	28.46	28.70	29.54
	0.8131	0.8298	0.8274	0.8109	0.8151	0.8400
Barbara	25.59	26.11	26.28	26.50	27.90	27.51
	0.7373	0.7580	0.7671	0.7621	0.8171	0.8078
Bike	24.24	24.38	24.11	24.35	24.77	25.39
	0.7588	0.7564	0.7509	0.7411	0.7740	0.8029
Leaves	26.53	26.66	26.37	27.15	27.57	28.28
	0.9036	0.8888	0.8904	0.9024	0.9088	0.9179